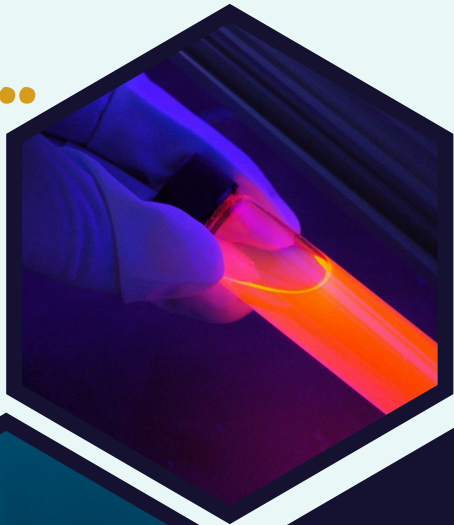


SSLC 2025

PART B

CHEMISTRY

learning material....



Sanjeevkumar Isaraddi

9448245781



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CHAPTER-1

CHEMICAL REACTIONS AND EQUATIONS

Chemical reaction

A chemical reaction is a process in which the original substance(s) loses its nature and identity and forms new substance(s) with different properties.

- The substances which take part in a chemical reaction are called **Reactants.**
- The substances which are formed in a chemical reaction are called **Products.**

A chemical reaction can be identified by either of the following observations:

(i) Change in state : The combustion reaction of candle wax is characterised by a change in state from solid to liquid and gas.

(ii) Change in colour : Shiny brown colored Copper on heating in air becomes a black colored compound.

$$2Cu + O_2 \rightarrow 2\ CuO$$

(iii) Evolution of gas : The chemical reaction between zinc and dilute sulphuric acid is characterised by hydrogen gas.

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$

(iv) Change in temperature : The reaction between quicklime and water to form slaked lime is characterised by an increase in temperature.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

(v) Formation of a precipitate : When an aqueous solution of sodium sulphate is mixed with the aqueous solution of barium chloride, barium sulphate comes in the form of white precipitate

$$Na_2SO_4\ (aq) + BaCl_2(aq) \rightarrow BaSO_4(\downarrow) + 2NaCl(aq)$$

Chemical equations

A chemical equation is the symbolic representation of a chemical reaction in the form of symbols and formulae. It is a way to represent the chemical reaction in a concise and informative way.

For example , Magnesium + Oxygen → Magnesium oxide
(Reactants) (Product)

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This equation is called word equation. The word equation can be written into chemical equation by writing symbols and formulae of the substance in place of their name.

$$2Mg + O_2 \rightarrow 2MgO$$

Writing a chemical equation

(i) The symbols of elements and the formulae of reacting substances (reactants) are written on the left hand side of the equation, with a plus (+) sign between them.

(ii) The symbols and formulae of the substances formed (products) are written on the right hand side of the equation, with a plus sign (+) between them.

(iii) An arrow sign (→) is put between the reactants and the products.

(iv) The physical states of the reactants and products are also mentioned in a chemical equation.

(v) The upward arrow sign (↑) denotes the evolution of gas and the downward arrow sign (↓) denotes the formation of precipitate

Balancing the chemical Equation:

The equation in which atoms of various elements on both sides of a chemical equation are equal in accordance with the law of conservation of mass.

Law of conservation of mass: It states that, "The matter can neither be created nor be destroyed in a chemical reaction.
The total mass of reactants = total mass of products.

The process of equalizing the atoms of various elements both on either sides of an equation is called the balancing of chemical equation.

Example 1 : Skeletal equation : $Mg + O_2 \rightarrow MgO$

Balanced equation : $2Mg + O_2 \rightarrow 2MgO$

Example 2 : Skeletal equation : $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

Balanced equation : $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Example 3 : Skeletal equation : $AgCl \rightarrow Ag + Cl_2$

Balanced equation : $2AgCl \rightarrow 2Ag + Cl_2$

Example 4 : Skeletal equation : $Al + O_2 \rightarrow Al_2O_3$

Balanced equation : $4Al + 3O_2 \rightarrow 2Al_2O_3$

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Types of Chemical Reactions

I. Combination Reaction:

The reaction in which two or more reactants combine to form a single product.

e.g., (i) Burning of coal $C + O_2 \rightarrow CO_2$

(ii) Formation of water $2H_2 + O_2 \rightarrow 2H_2O$

(iii) $CaO + H_2O \rightarrow Ca(OH)_2 + Heat$
(Quick lime) (Slaked lime)

Exothermic Reactions : Reaction in which heat is released along with formation of products.

e.g.,(i) Burning of natural gas.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + Heat$

(ii) Respiration is also an exothermic reaction.

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + energy$

II. Decomposition Reaction:

The reaction in which a compound splits into two or more simpler substances is called decomposition reaction. $A \rightarrow B + C$

(a) Thermal decomposition: When decomposition is carried out by heating.

e.g., (i) $2FeSO_4 \xrightarrow{Heat} Fe_2O_3 + SO_2 + SO_3$
(Ferrous sulphate) Green colour — (Ferric oxide) Red-brown colour

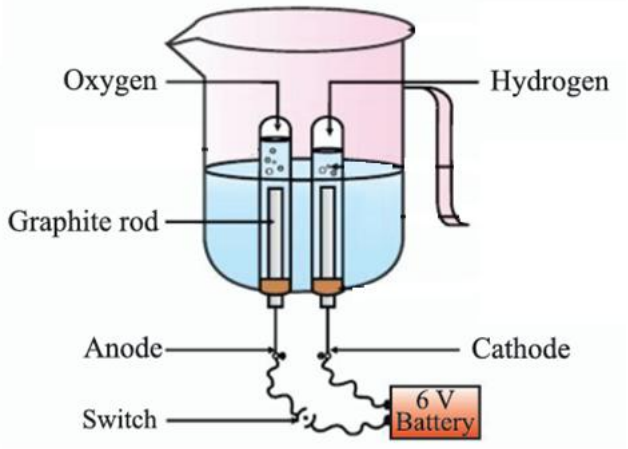
(ii) $CaCO_3 \xrightarrow{Heat} CaO + CO_2$
(Lime stone) (Quick lime)

(iii) $2Pb(NO_3)_2 \xrightarrow{Heat} 2PbO + 4NO_2 + O_2$
(Lead nitrate) (Lead oxide) (Nitrogen dioxide) (Brown fumes)

(b) Electrolytic Decomposition: When decomposition is carried out by passing electricity. e.g., Electrolysis of water

$$2H_2O \xrightarrow{Electric\ current} 2H_2 + O_2$$

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- Pure water is a bad conductor of electricity. So to carryout electrolysis it is essential to add few drops of dil. H_2SO_4 ,before the electrolysis
- Gas collected at cathode is Hydrogen and at anode is Oxygen
- Water contains two parts of hydrogen element as compared to one part of oxygen element by volume. Hence, volume of gas collected at one electrode is double of another

(c) Photolytic Decomposition: When decomposition is carried out in presence of sunlight.

e.g., (i) White silver chloride turns grey in sunlight. This is due to the decomposition of silver chloride into silver and chlorine by light.

$$\underset{\text{(White)}}{2AgCl} \xrightarrow{Sunlight} \underset{\text{(Grey)}}{2Ag} + Cl_2$$

(ii) $2AgBr \xrightarrow{Sunlight} 2Ag + Br_2$

These reactions are used in black and white photography.

Endothermic Reaction: The reactions which require energy in the form of heat, light or electricity to break reactants are called endothermic reactions.

III. Displacement Reaction: The chemical reactions in which more reactive element displaces less reactive element from its salt solution.

e.g., (i) $\underset{\text{(Iron)}}{Fe} + \underset{\text{(Copper sulphate) (Blue)}}{CuSO_4} \rightarrow \underset{\text{(Ferrous sulphate) (Green)}}{FeSO_4} + \underset{\text{(Copper)}}{Cu}$

(ii) Zinc displaces copper forming zinc sulphate. Zn is more reactive than copper.

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$$

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IV. Double Displacement Reaction: A reaction in which new compounds are formed by mutual exchange of ions between two compounds.

e.g., $Na_2SO_4(aq)$ + $BaCl_2(aq)$ → $BaSO_4(s)$ + $2NaCl(aq)$
(Sodium sulphate) (Barium chloride) (Barium sulphate) (Sodium chloride)

White precipitate of $BaSO_4$ is formed, so it is also called precipitation reaction.

Other eg, : $Al_2(SO_4)_3 + 3BaCl_2 \rightarrow 2AlCl_3 + 3BaSO_4\downarrow$

V. Oxidation and Reduction:

Oxidation: Loss of electrons Reduction: Gain of electrons

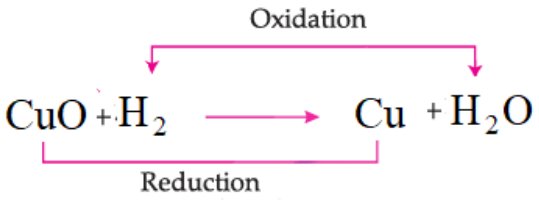
Oxidation: It is a process of gaining oxygen during a reaction by an atom, molecule or ion.

$$2Cu + O_2 \xrightarrow{Heat} 2CuO$$

Reduction: It is the gain of electrons or a decrease in the oxidation state of an atom by another atom, an ion or a molecule.

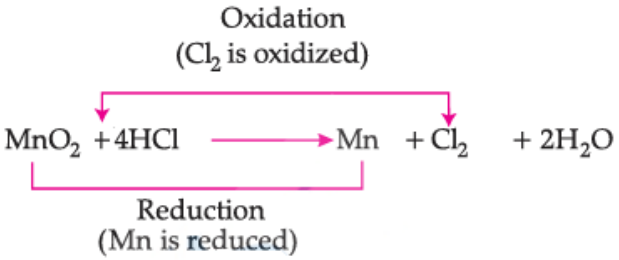
$$CuO + H_2 \rightarrow Cu + H_2O$$

In this reaction, CuO is reduced to Cu and H_2 is oxidised to H_2O.



In other words, one reactant gets oxidised while the other gets reduced. Such reactions are called oxidation-reduction reactions or redox reactions.

e.g.,



Effects of oxidation reactions in everyday life :

1. Corrosion: Corrosion is a process in which metals are deteriorated by action of air, moisture, chemicals, etc. It is a redox reaction where metal gets oxidised to metal oxide and oxygen gets reduced to oxide ion.

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Examples: (a) Corrosion of iron is called rusting. Iron objects when left in moist open air for sometime get coated with a reddish brown powder. The process is known as rusting.

(b) Green coating on copper articles and black coating on silver ornaments are other examples of corrosion.

Prevention of Rusting :

(a) The iron articles should be painted.

(b) The machine parts should be oiled and greased.

(c) Galvanised iron pipes should be used for water supply.

(d) Iron can be coated with chromium to prevent rusting.

2. Rancidity : Rancidity is the process of slow oxidation of oil and fat, present in the food materials resulting in the production of foul odour and taste in them. When cooked food items are placed for a long time, they become rancid and unsuitable for the consumption.

Rancidity can be prevented by the following ways:

(a) Storing the food in refrigerator.

(b) Storing the food in air-tight container.

(c) Addition of anti-oxidants to food.

(d) Storing the food in flush bags with gas, such as nitrogen to prevent the oxidation process.

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CHAPTER-2

ACIDS, BASES AND SALTS

Acids

- Acids are the substances that furnish H+ ions in aqueous solution.
- Acids are sour in taste.
- They turn blue litmus red.
- If in an aqueous solution, concentration of acid is low, it is called **dilute** and if concentration of acid is high, it is called **concentrated**.
- Those acids which dissociates into ions completely are called strong acids, **e.g.,** H_2SO_4, HCl. HNO_3
- Those acids which do not dissociate into ions completely are called weak acids, e.g., citric acid, acetic acid.

Bases

- Bases are those chemical compounds which are bitter in taste, soapy in touch,
- Turn red litmus blue
- Give OH– ions in aqueous solution.
- The bases which ionise completely to furnish OH^- ions are called strong bases, e.g., KOH, NaOH, etc.
- The bases which ionise only partially are called weak bases, e.g., $Mg(OH)_2$, $Cu(OH)_2$, etc.

Indicators:

These are the substances which change their colour / smell in different types of substances.

	Red litmus	Blue litmus	Phenolphthalein	Methyl orange
Acidic solution	No change in color	Turns in to red color	No change in color	It turns orange to red color
Basic solution	Turns in to blue color	No change in color	It turns to pink	It turns orange to yellow color

There are some substances whose odour changes in acidic or basic media. These are called **olfactory indicators.** Eg, vanilla, onion and clove

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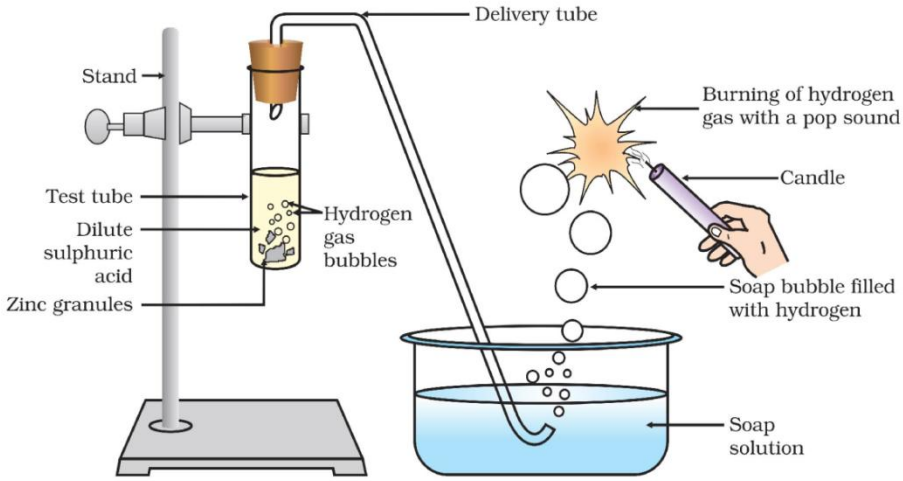
Chemical Properties of Acids and Bases

1. Reaction with Metals

Acids	Bases
Acid + Metal → Salt + Hydrogen gas	Acid + Metal → Salt + Hydrogen gas
e.g., $2HCl + Zn \rightarrow ZnCl_2 + H_2 \uparrow$	e.g., $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2 \uparrow$ (Sodium zincate)

Test for H_2 gas : Hydrogen gas released can be tested by bringing a burning candle near gas bubbles, it bursts with pop sound.

Reaction of zinc granules with dilute sulphuric acid and testing hydrogen gas by burning



2. Reaction with Metal carbonates and hydrogen carbonates

All metal carbonates and hydrogencarbonates react with acids to give a corresponding salt, carbon dioxide and water.

Eg, (i) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2 \uparrow$
Sodium carbonate

(ii) $NaHCO_3 + 2HCl \rightarrow NaCl + H_2O + CO_2 \uparrow$
Sodium hydrogen carbonate

Test for CO_2 : CO_2 can be tested by passing it through lime water. Lime water turns milky.

$$Ca(OH)_2 + CO_2 \rightarrow \underset{\text{Soluble salt}}{CaCO_3} + H_2O$$

When excess CO_2 is passed, milkiness disappears.

$$CaCO_3 + CO_2 + H_2O \rightarrow \underset{\text{Insoluble salt}}{Ca(HCO_3)_2}$$

Bases do not react with metal carbonates and hydrogencarbonates

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3. Reaction of Acids and Bases With Each Other :

When an acid reacts with base, the hydrogen ion of acid combines with hydroxide ion of base and forms water. As these ions combine together, they form water instead of remaining free, thus both neutralize each other.

$$\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$$

Since, in the reaction between an acid and a base both neutralize each other, it is also known as neutralization reaction.

Example : Sodium hydroxide (a strong base) reacts with hydrochloric acid to form sodium chloride and water.

$$NaOH + HCl \rightarrow NaCl + H_2O$$

4. Reaction of acids with metal oxides :

Metal oxides react with acids to give salt and water.

$$\text{Metal oxide} + \text{Acid} \rightarrow \text{Salt} + \text{Water}$$

Example : Copper oxide reacts with dil. hydrochloric acid to form copper chloride (salt) and water.

$$\underset{\text{Copper oxide}}{CuO} + 2HCl \rightarrow \underset{\text{Copper chloride}}{CuCl_2} + H_2O$$

Copper oxide is black in colour. When dilute hydrochloric acid is added in it, the colour of the solution becomes blue green due to formation of copper chloride.

Strength of an acid or base

- Strength of an acid or base depends on the number of H+ ions or OH– ions produced by them respectively.
- Based on its ability to dissociate into ions in solution, acids and bases are classified as strong acid or base and weak acid or base.

(i) Strong acids : Acid which completely dissociates in water to produce large amount of hydrogen ions are called strong acids. For example hydrochloric acid (HCl), sulphuric acid (H_2SO_4), nitric acid (HNO_3) are strong acids as they get completely ionized in water to form ions.

$$HCl\ (aq) \rightarrow H^+(aq) + Cl^-\ (aq)$$

(ii) Weak acids : Acids which get partially ionized in water to produce small amount of hydrogen ions are known as weak acids. For example, acetic acid partially dissociates in water to produce small amount of hydrogen ions.

$$CH_3COOH\ (aq) \rightarrow H^+\ (aq) + CH_3COO^-(aq)$$

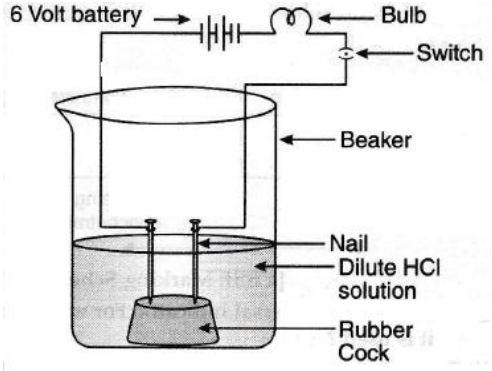
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(iii) Strong bases : Bases which completely ionize in water to produce large amount of hydroxide ions are called strong bases. Ex : NaOH, KOH, etc.

(iv) Weak bases : Bases which partially dissociate in water to furnish lesser amount of hydroxide ions are called weak bases. Ex: Ammonium hydroxide (NH_4OH) and calcium hydroxide $Ca(OH)_2$.

Conduction of electricity

Take solutions of glucose, alcohol, hydrochloric acid, sodium hydroxide, etc. Fix two nails on a cork, and place the cork in a 100 mL beaker. Connect the nails to the two terminals of a 6 volt battery through a bulb and a switch. Now pour solution to be tested in the beaker and switch on the current.



Observations :

- The bulb will glow in acidic and basic solution as they produce H+ or OH- ions.
- The bulb will not glow in alcohol or glucose solution as they do not produce ions and hence, they do not conduct electricity

Dilution of Acids

Mixing an acid or base with water results in decrease in the concentration of ions (H_3O^+/OH^-) per unit volume. Such a process is called dilution

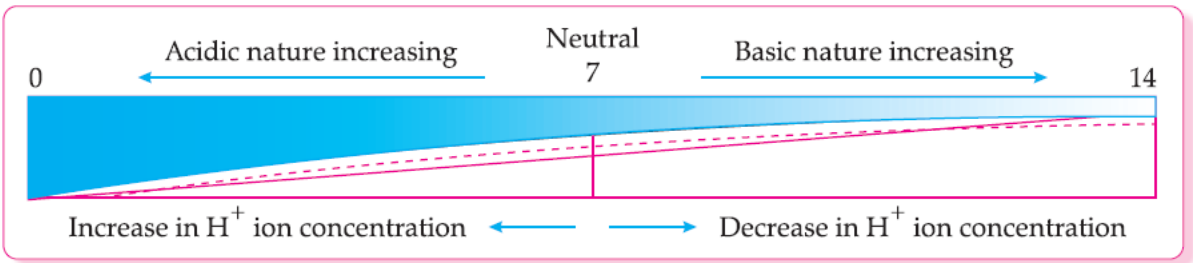
- The process of dissolving an acid or a base in water is a highly exothermic one. Care must be taken while mixing concentrated acid with water.
- The acid must always be added slowly to water with constant stirring.
- If water is added to a concentrated acid, the heat generated may cause the mixture to splash out and cause burns. The glass container may also break due to excessive heat.

pH scale :

- A scale for measuring hydrogen ion concentration in a solution, is called pH scale.
- On the pH scale we can measure pH generally from 0 (very acidic) to 14 (very alkaline).

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- The pH of a neutral solution is 7.
- Values less than 7 on the pH scale represent an acidic solution. It means pH 5 is more acidic than pH 6
- As the pH value increases from 7 to 14, it represents an increase in OH– ion concentration in the solution, that is, increase in the strength of base



Difference between litmus paper and pH paper :

- Litmus paper can be used to distinguish between an acid and a base.
- pH paper is used to measure the hydrogen ion concentration in a solution or strength of an acid and a base.

Importance of pH in Everyday Life

- Plants and animals are pH sensitive.
- Our body works within the pH range of 7 – 7.8.
- When pH of rain water is less than 5.6, it is called acid rain. Acid rain is rain or any other form of precipitation that is unusually acidic, meaning that it has elevated levels of hydrogen ions (low pH).
- Plants require a specific pH range for their healthy growth.
- pH of stomach is 1.5-3.0 due to secretion of HCl. In case of indigestion, acidity increases, which can be neutralised by antacids like milk of magnesia.
- Tooth decay starts when pH of the mouth is lower than 5.5. To protect tooth decay, toothpastes which are basic in nature are used to neutralize the excess acid.
- Many plants and animals produce certain acids to defend themselves. For example, Bee stings leave an acid into the skin, which causes pain and irritation. If a mild base like baking soda is applied on the stung area, it gives relief.

Family of Salts

(i) Neutral salts : Salts produced by reaction of strong acid and strong base are neutral in nature with pH value 7. For example, sodium chloride formed by reaction between sodium hydroxide (strong base) and hydrochloric acid (strong acid).

$$NaOH\ (aq) + HCl\ (aq) \rightarrow NaCl\ (aq) + H_2O\ (l)$$

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(ii) Acidic salts : Salts formed by reaction between a strong acid and weak base are acidic in nature with pH value less than 7. For example, ammonium hydroxide. It is a salt of hydrochloric acid (strong acid) and ammonium hydroxide (weak base).

$$NH_4OH\ (aq) + HCl(aq) \rightarrow NH_4Cl\ (aq) + H_2O\ (l)$$

(iii) Basic salts : Salts formed by reaction of strong base and weak acid are basic in nature with pH value more than 7. For example, sodium carbonates. It is a salt of carbonic acid (weak acid) and sodium hydroxide (strong base).

$$H_2CO_3\ (aq) + 2NaOH\ (aq) \rightarrow Na_2CO_3\ (aq) + 2H_2O\ (l)$$

Common salt — A raw material for chemicals

The common salt thus obtained is an important raw material for various materials of daily use, such as sodium hydroxide, baking soda, washing soda, bleaching powder and many more.

Sodium hydroxide

- When electricity is passed through an aqueous solution of sodium chloride (brine solution), it decomposes to form sodium hydroxide. The process is called the chlor-alkali process.
- In this process, Chlorine gas is given off at the anode, and hydrogen gas at the cathode.
- Sodium hydroxide solution is formed near the cathode.
- The three products produced in this process are all useful.

Uses :

(a) Sodium hydroxide is used in making of paper, soap and detergents, for de-greasing metals, etc.

(b) Chlorine gas is used in water treatment, manufacturing of PVC, pesticides, etc.

(c) Liquid hydrogen is used as rocket fuel, in hydrogenation process of oil to produce vegetable ghee (margarine) and in making of ammonia for fertilizers.

Bleaching powder

Chemical name : Calcium oxychloride

Chemical formula : $CaOCl_2$

Preparation : It is produced by the action of chlorine on dry slaked lime.

$$Cl_2 + Ca(OH)_2 \rightarrow CaOCl_2 + H_2O$$

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Uses: (a) Bleaching cotton and linen in textile industry.

(b) Bleaching wood pulp in paper factories.

(c) Oxidizing agent in chemical industries.

(d) Disinfecting drinking water.

Baking soda

Chemical name : Sodium hydrogen carbonate

Chemical formula : $NaHCO_3$

Preparation : $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$

Uses : (a) For making baking powder (mixture of baking soda and tartaric acid).

(b) An ingredient in antacid.

(c) Used in soda acids, fire extinguishers.

Washing soda

Chemical name : Sodium carbonate

Chemical formula : $NaCO_3 \cdot 10H_2O$

Preparation : Re-crystallization of sodium carbonate gives washing soda.

It is a basic salt.

$$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3.10H_2O$$

Uses : (a) In glass, soap and paper industry.

(b) Manufacture of borax.

(c) It can be used as cleaning agent.

(d) It can be used for removing permanent hardness of water.

Plaster of Paris

Chemical name : Calcium sulphate hemihydrate

Chemical formula : $CaSO_4 \cdot \frac{1}{2}H_2O$

Preparation : On heating gypsum CaSO4.$2H_2O$ at 373K, it loses water molecules and becomes Plaster of Paris (POP). It is white powder and on mixing with water it changes to gypsum.

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$$CaSO_4 \cdot 2H_2O \xrightarrow{373K} CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$$

Gypsum — Plaster of Paris

$$CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O \rightarrow CaSO_4 \cdot 2H_2O$$

Plaster of Paris — Gypsum

Uses: (a) Doctors use POP for supporting fractured bones.

(b) For making toys and material for decoration.

Water of crystallization

Water molecules present in the crystal structure of salt are called water of crystallization and such salts are called hydrated salts. Water of crystallization is the fixed number of water molecule present in one formula unit of a salt.

Examples:

- Copper sulphate pentahydrate ($CuSO_4.5H_2O$) : It has five water molecules in one formula unit of copper sulphate (blue vitriol).
- Sodium carbonate ($Na_2CO_3.10H_2O$) : It has ten molecules of water as water of crystallization.
- Gypsum ($CaSO_4.2H_2O$) : It has two molecules of water as water of crystallization.

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CHAPTER-3

METALS AND NON-METALS

Physical Properties of Metals and Non-metals

- Metals, in their pure state, have a shining surface. This property is called metallic luster.
- Metals are generally hard. The hardness varies from metal to metal. Except Sodium, Lithium and Potassium which are soft and can be easily cut with knife.
- Metals can be beaten into thin sheets. This property is called malleability. (Gold and silver are the most malleable metals)
- The ability of metals to be drawn into thin wires is called ductility. Gold is the most ductile metal.
- Metals are good conductors of heat and have high melting points. This is why metals are used in cooking vessels. (Lead and mercury are comparatively poor conductors of heat)
- Metals are good conductors of electricity.
- The metals that produce a sound on striking a hard surface are said to be sonorous.

Some exceptions :

- Mercury exist as liquid at room temperature.
- Gallium and caesium have very low melting points. These two metals will melt if you keep them on your palm.
- Iodine is a non-metal but it is lustrous.
- Diamond, an allotrope of carbon, is the hardest natural substance known and has a very high melting and boiling point. Graphite, another allotrope of carbon, is a conductor of electricity.

Chemical Properties of Metals

Reaction of Metals with Air :

Metals combine with oxygen to form metal oxide.

$$\text{Metals} + O2 \rightarrow \text{Metal oxide}$$

Examples: (i) $2Cu + O_2 \rightarrow 2CuO$

Copper (II)oxide (black)

(ii) $4Al + 3O_2 \rightarrow 2Al_2O_3$

Aluminium oxide

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(iii) $2Mg + O_2 \rightarrow 2MgO$
Magnesium oxide

Different metals show different reactivity towards O_2.

- Na and K react so vigorously with oxygen that they catch fire if kept in open. So, they are kept immersed in kerosene.
- Surfaces of Mg, Al, Zn and Pb are covered with a thin layer of oxide which prevent them from further oxidation.
- Fe does not burn on heating but iron fillings burn vigorously.
- Au and Ag do not react with oxygen.

Amphoteric Oxides :

Metal oxides which react with both acids as well as bases to produce salt and water are called amphoteric oxides. Eg , Aluminium oxide and Zinc Oxide

$$Al_2O_3 + 6HCl \rightarrow \underset{\text{Aluminium chloride}}{2AlCl_3} + 3H_2O$$

$$Al_2O_3 + 2NaOH \rightarrow \underset{\text{Sodium aluminate}}{2NaAlO_2} + H_2O$$

Reaction of Metals with Water :

Metals react with water to produce metal hydroxide and hydrogen gas.

Metal + Water → Metal oxide + Hydrogen

Examples: $2Mg + 2H_2O \rightarrow \underset{\text{Magnesium oxide}}{2MgO} + 2H_2\uparrow$

Metal oxides when dissolves in water produce Metal hydroxides, Hence, metal oxides are said to be basic in nature

Metal oxide + Water → Metal hydroxide (Base)

$$MgO + H_2O \rightarrow \underset{\text{Magnesium hydroxide}}{Mg(OH)_2}$$

Non-metal oxides dissolves in water produce acids, Hence, non-metal oxides are said to be acidic in nature

For example, sulphur trioxide (non-metal oxide) reacts with water to form sulphuric acid.

$$SO_3 + H_2O \rightarrow \underset{\text{sulphuric acid}}{H_2SO_4}$$

- Sodium and Potassium react vigorously with water. the reaction is so violent and exothermic that the evolved hydrogen immediately catches fire.

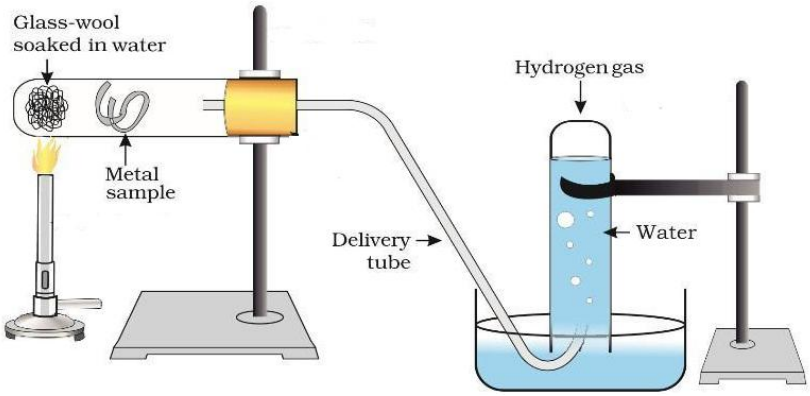
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- The reaction of calcium with water is less violent. The heat evolved is not sufficient for the hydrogen to catch fire.

$$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$$

- Magnesium metal reacts with hot water to produce magnesium hydroxide and hydrogen gas. It starts floating due to the bubbles of hydrogen gas sticking to its surface.
- Aluminium, iron and zinc react with steam to produce metal oxide and hydrogen gas.
- Metals like silver, gold, copper and lead do not react with water.

Action of steam on a metal



Reactions of Metals with Acid:

$$\text{Metal} + \text{Dil. Acid} \rightarrow \text{Salt} + \text{Hydrogen gas}$$

$$\text{e.g., } Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$$

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2$$

- Copper, mercury and silver don't react with dilute acids.
- Hydrogen gas is not evolved when a metal reacts with nitric acid. It is because HNO_3 is a strong oxidising agent. It oxidises the H_2 produced to water and itself gets reduced to any of the nitrogen oxides (N_2O, NO, NO_2).
- When metals react with Dilute HCl, the rate of formation of hydrogen bubbles is different in each case.It was fastest in case of magnesium and in the case of copper, no bubbles were seen

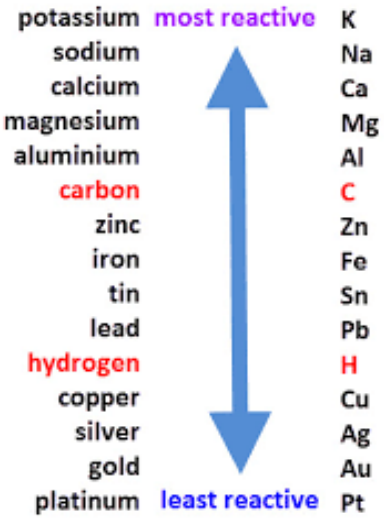
Metals react with Solutions of other Metal Salts

- Reactive metals can displace less reactive metals from their compounds in solution or molten form.
- If metal A displaces metal B from its solution, it is more reactive than B.

Metal A + Salt solution of B → Salt solution of A + Metal B

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Reactivity series of Metals :



Reaction between metal and non-metals:

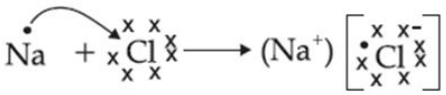
- Reactivity of an element is the tendency to attain completely filled valence shells.
- Atoms of metals can lose electrons from valence shells to form cations while atoms of non-metals can gain electrons in valence shell to form anions.
- Opposite charged ions attract each other and held by strong electrostatic forces of attraction.

Formation of NaCl :

- Atomic number of sodium is 11. Electron configuration is 2, 8 , 1. Electron in last shell is 1.
- Atomic number of chlorine is 17. Electron configuration is 2, 8 , 7. Electrons in last shell are 7.
- Sodium gives 1 electron to Chlorine atom to attain octave configuration.
- By loosing an electron sodium will gets a unit positive charge and becomes a cation , while by gaining an electron chlorine gets a unit negative charge and becomes anion.
- Sodium and chloride ions, being oppositely charged, attract each other and are held by strong electrostatic forces of attraction called ionic bond.

$$\underset{2,8,1}{Na} \rightarrow \underset{2,8}{Na^+} + e^- \quad \text{(Sodium cation)}$$

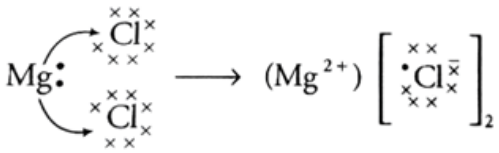
$$\underset{2,8,7}{Cl} + e^- \rightarrow \underset{2,8,8}{Cl^-} \quad \text{(Chloride anion)}$$



Formation of $MgCl_2$:

$$\underset{(2,8,2)}{Mg} \longrightarrow \underset{(2,\ 8)}{Mg^{2+}} + 2e^- \quad \text{(Magnesium cation)}$$

$$\underset{(2,8,7)}{Cl} + e^- \longrightarrow \underset{(2,\ 8,\ 8)}{Cl^-} \quad \text{(Chloride anion)}$$



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Ionic Compounds

The compounds formed by the transfer of electrons from a metal to a non-metal are called ionic compounds or electrovalent compounds.

Properties of Ionic Compounds :

(i) Physical nature: They are solid and hard, generally brittle. (because of the strong force of attraction between the positive and negative ions.)

(ii) Melting and Boiling Point: They have high melting and boiling points. This is because a considerable amount of energy is required to break the strong inter-ionic attraction.

(iii) Solubility: Generally soluble in water and insoluble in solvents such as kerosene, petrol, etc.

(iv) Conduction of electricity: Ionic compounds conduct electricity in molten and solution form but not in solid state.(because, in solutions ions move freely and conducts electricity)

Extraction of metals

Metallurgy : The extraction of metals from their ores and then refining them for use is known as metallurgy.

Minerals : The elements or compounds which occur naturally in the earth's crust are called minerals.

Ores : Minerals that contain very high percentage of particular metal and the metal can be profitably extracted from it, such minerals are called ores.

Extraction of metals : It is the process of obtaining pure metal from its ore. Extraction of metal can be classified into three steps :

- Enrichment of ores or concentration of ores.
- Extraction of metal from the concentrated ores.
- Refining of metal.

On the basis of reactivity, metals can be grouped into three categories:

(i) Metals of low reactivity
(ii) Metals of medium reactivity
(iii) Metals of high reactivity.

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Metals of low reactivity

- Metals at the bottom of the activity series are least reactive and are often found in free state. e.g., Gold, silver, platinum and copper.
- These metals are very least reactive. The oxides of these metals can be reduced to metals by heating alone.
- For example, cinnabar (HgS) (an ore of mercury). When it is heated in air, it is first converted into mercuric oxide which is further reduced to mercury on heating.

$$2HgS + 3O_2 \xrightarrow{Heat} 2HgO + 2SO_2$$

$$2HgO \xrightarrow{Heat} 2Hg + O_2$$

- Similarly, copper which is found as Cu2S in nature can be obtained from its ore by just heating in air.

$$2Cu_2S + 3O_2 \xrightarrow{Heat} 2Cu_2O + 2SO_2$$

$$2Cu_2O + Cu_2S \xrightarrow{Heat} 6Cu + SO_2$$

Metals of middle reactivity

- The metals in the middle of the activty series such as iron, zinc, lead, copper, are moderately reactive. These are usually present as sulphides or carbonates in nature.
- It is easier to obtain a metal from its oxide, as compared to its sulphides and carbonates.
- Therefore, prior to reduction, the metal sulphides and carbonates must be converted into metal oxides.
- The sulphide ores are converted into oxides by heating strongly in the presence of excess air. This process is known as **roasting.**
- The carbonate ores are changed into oxides by heating strongly in limited air. This process is known as **calcination.**

Example : Extraction of Zinc

Roasting : $2ZnS + 3O_2 \xrightarrow{Heat} 2ZnO + 2SO_2$

Calcination : $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$

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Reduction : The metal oxides are then reduced by using suitable reducing agents such as carbon. For example, when zinc oxide is heated with carbon, it is reduced to metallic zinc.

$$ZnO + C \xrightarrow{Heat} Zn + CO$$

The highly reactive metals such as sodium, calcium, aluminium, etc., are used as reducing agents because they can displace metals of lower reactivity from their compounds.

These displacement reactions are highly exothermic. The amount of heat evolved is so large that the metals are produced in the molten state.

For example, the reaction of iron(III) oxide (Fe_2O_3) with aluminium is used to join railway tracks or cracked machine parts. This reaction is known as the **thermit reaction.**

$$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + Heat$$

Metals of high reactivity

- Sodium, Potassium, Magnesium, Calcium, Aluminium, etc., are the highly reactive metals.
- They can not be reduced by reducing agents like carbon, because these metals have more affinity for oxygen than carbon.
- These metals are obtained by electrolytic reduction.
- For example, sodium, magnesium and calcium are obtained by the electrolysis of their molten chlorides. The metals are deposited at the cathode (the negatively charged electrode), whereas, chlorine is liberated at the anode (the positively charged electrode).

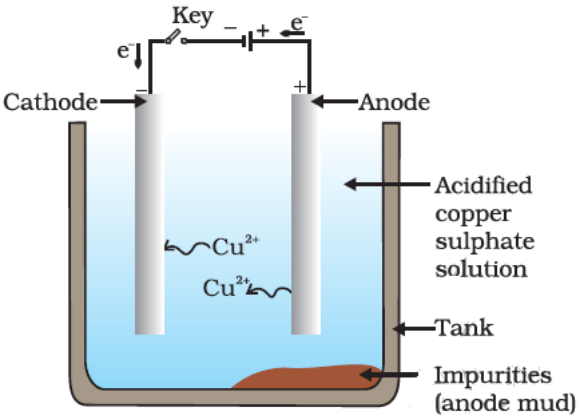
Refining of Metals

The metals produced by various reduction processes described above are not very pure. They contain impurities, which must be removed to obtain pure metals. The most widely used method for refining impure metals is electrolytic refining.

Electrolytic refining.

- Many metals, such as copper, zinc, tin, nickel, silver, gold, etc., are refined electrolytically. In this process, the impure metal is made the anode and a thin strip of pure metal is made the cathode. A solution of the metal salt is used as an electrolyte.
- On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte. An equivalent amount of pure metal from the electrolyte is deposited on the cathode. The soluble impurities go into the solution, whereas, the insoluble impurities settle down at the bottom of the anode and are known as **anode mud.**

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Corrosion:

It is the deterioration of a metal as a result of chemical reactions between it and surrounding environment.

For example,

- Silver reacts with sulphur in air to form silver sulphide and articles become black.
- Copper reacts with moist carbon dioxide in air and forms green coat of copper carbonate.
- Iron acquires a coating of brown flaky substance called rust.
- Rust is hydrated Iron (III) oxide, i.e., $Fe_2O_3.xH_2O$

Prevention of corrosion:

By painting, oiling, greasing, galvanizing and by making alloys.

Galvanization: It is the process which involves coating of iron with zinc. The oxide thus formed is impervious to air and moisture thus protects further layers from getting corroded.

Alloys

These are homogeneous mixture of metals with metals and non-metals.

For example,

- Stainless steel: Alloy of iron, nickel, chromium
- Brass : Alloy of copper and zinc
- Bronze : Alloy of copper and tin
- Solder : Alloy of lead and tin has a low melting point and is used for welding electrical wires together.

Amalgam: If one of the metals is mercury, then the alloy is known as amalgam, e.g., sodium amalgam and silver amalgam.

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CHAPTER-4

CARBON AND ITS COMPOUNDS

Properties of Carbon

- Carbon is **tetravalent** in nature.
- It does not form ionic bond because it has 4 valence electrons.
- To form ionic bonds, carbon molecules must either gain or lose 4 electrons.
- It is difficult to hold four extra electrons in its outer shell hence, it will not form $\boldsymbol{C^{-4}}$ ***ion*** and would require large amount of energy to remove four electrons. So, it will not form $\boldsymbol{C^{+4}}$ ***ion***
- So, carbon can form bond by sharing of its electron with the electrons of other carbon atom or with other element to attain noble gas configuration.

Covalent bond

The bond formed by sharing of electrons between same or different atoms is called **covalent bond**.
In a covalent bond, shared pair of electrons belongs to the valence shell of both atoms.
Examples :

The atomic number of hydrogen is 1. Hence hydrogen has one electron in its K shell and it requires one more electron to fill the K shell. So two hydrogen atoms share their electrons to form a molecule of hydrogen, H_2.

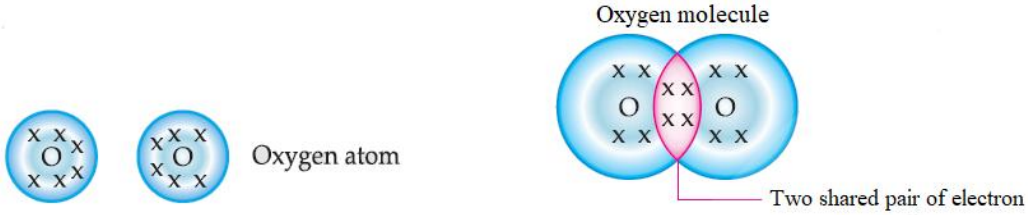
The shared pair of electrons is said to constitute a single covalent bond between the two hydrogen atoms.



H – H Single bond between hydrogen atoms

The atomic number of oxygen is 8, it has 6 electrons in its L shell it requires two more electrons to complete its octet. So each atom of oxygen shares two electrons with another atom of oxygen

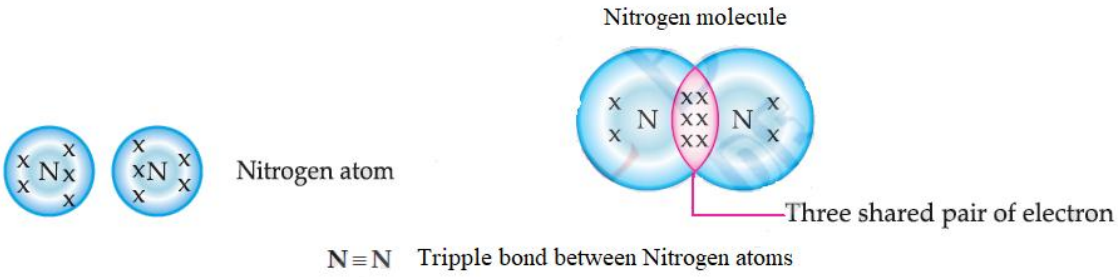
The two electrons contributed by each oxygen atom give rise to two shared pairs of electrons. This is said to constitute a double bond between the two atoms.



O = O Double bond between oxygen atoms

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Nitrogen has the atomic number 7. In order to attain an octet, each nitrogen atom in a molecule of nitrogen contributes three electrons giving rise to three shared pairs of electrons. This is said to constitute a triple bond between the two atoms.



Properties of covalent compounds:

(i) Physical state: The covalent compounds exist as gases liquids and soft solids.

(ii) Solubility: They are generally insoluble in water and other polar solvents but soluble in organic solvents such as benzene, toluene, etc.

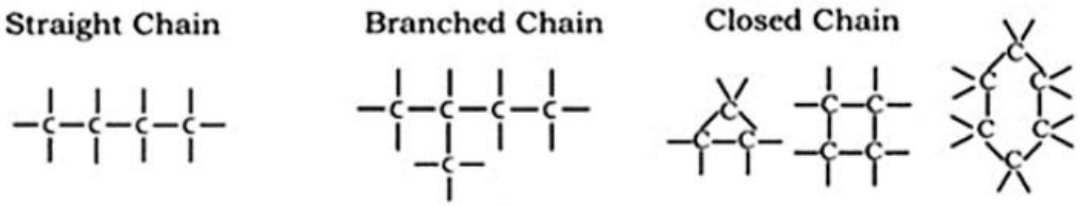
(iii) Melting and boiling points: They generally have low melting and boiling points. The molecules of covalent compounds are held by weak intra-molecular forces. Thus, a very small amount of energy is required to break the bonds between two or more molecules. That is why they have low melting and boiling points.

(iv) Electrical conductivity : They are non-conductors of electricity in solid, molten or aqueous state. As covalent bonds lack free ions, they are not able to conduct electricity.

Versatile Nature of Carbon

Carbon can form large number of carbon compounds. The factors that enable carbon to form large number of compounds are catenation and tetravalency.

Catenation: It is the unique ability of elements to form long, straight or branched chains and rings of different sizes.

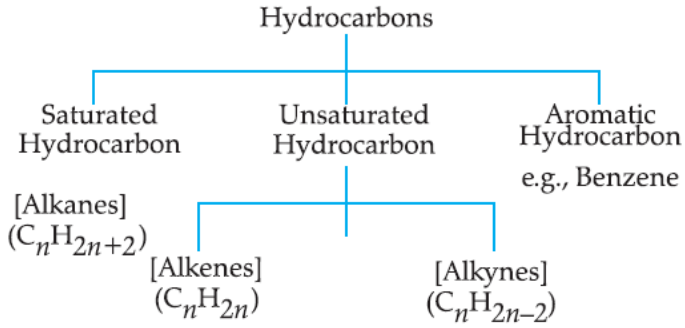


Tetravalency : It is the state of an atom in which there are four electrons available with the atom for covalent chemical bonding.

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Hydrocarbons

Compounds made up of hydrogen and carbon are called hydrocarbons.



IUPAC name of hydrocarbon consists of two parts.

(i) Word root: Number of carbons in the carbon chain.

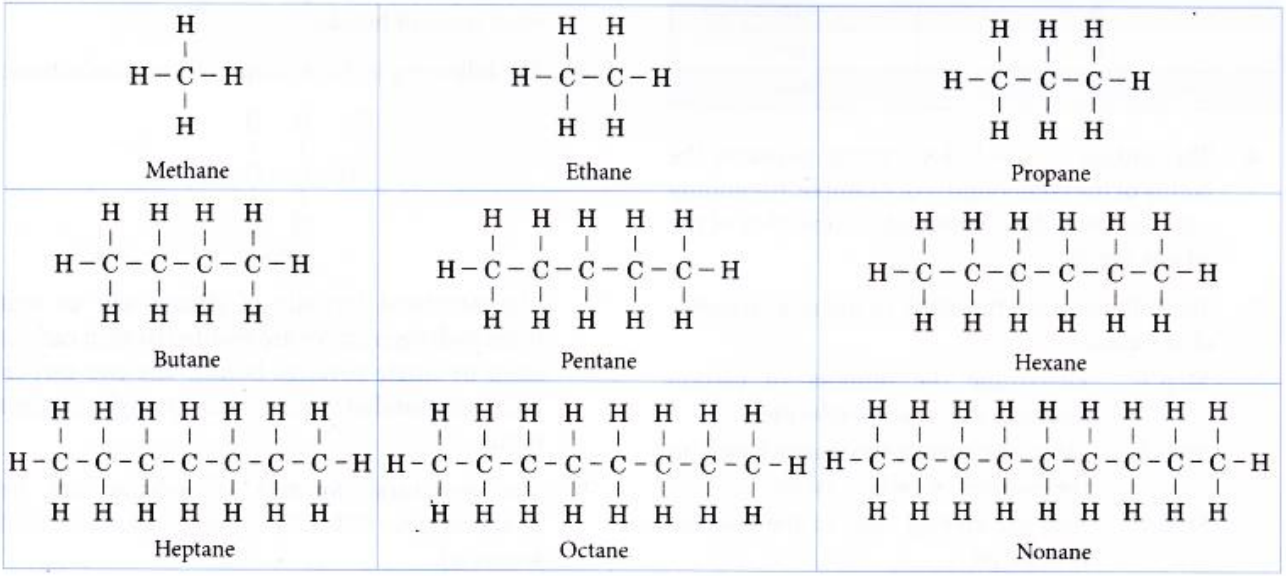
Number of carbon atoms	Word root (Greek name)
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex
7	Hept
8	Oct
9	Non
10	Dec

(ii) Suffix: It depends on the type of carbon - carbon bond.

For single bond suffix is – ane , for double bond, suffix is – ene , and for triple bond suffix is – yne.

Homologous series of alkanes

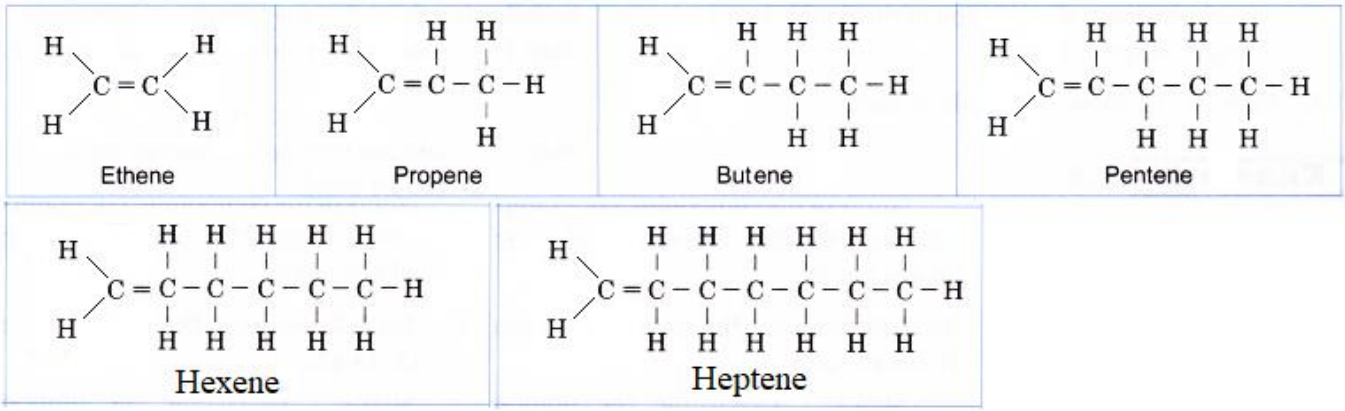
General formula: C_nH_{2n+2}, where n = number of carbon atoms.



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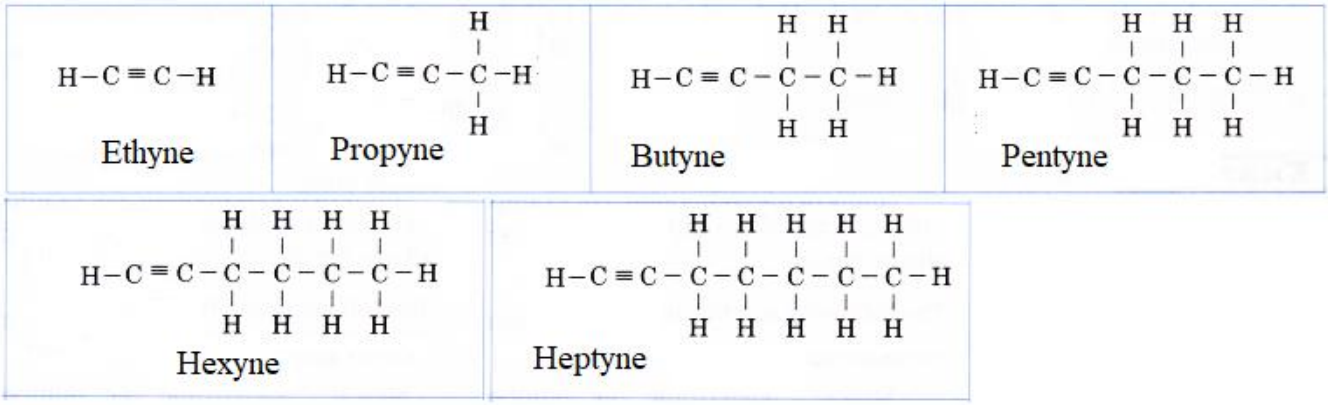
Homologous series of alkenes

General formula: C_nH_{2n} , where n = number of carbon atoms.



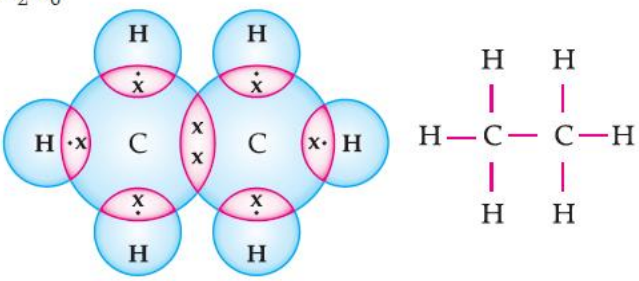
Homologous series of alkynes

General formula: C_nH_{2n-2}, where n = number of carbon atoms.



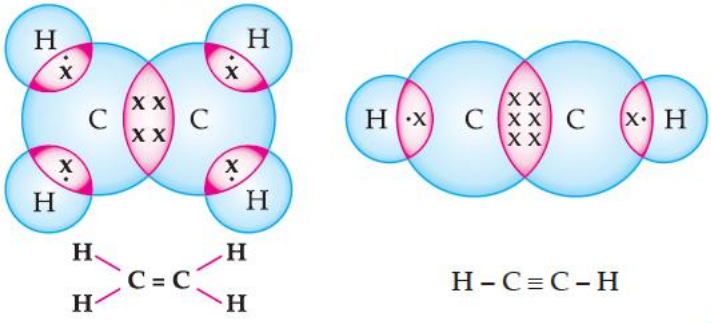
Electron dot structure of saturated hydrocarbons:

Ethane C_2H_6



Electron dot structure of unsaturated hydrocarbons:

Ethene: C_2H_4 Ethyne: C_2H_2

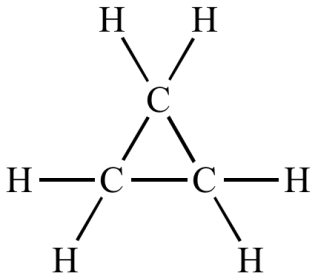


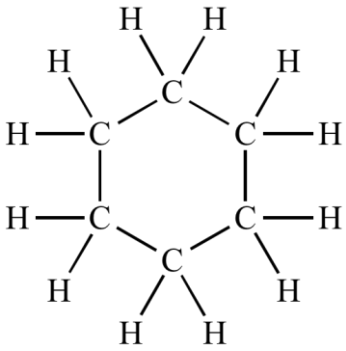
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Cyclic or Closed Chain Hydrocarbons:

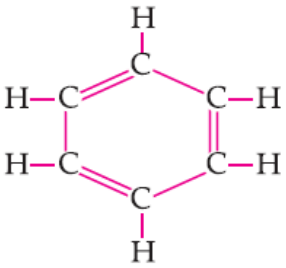
These are the hydrocarbons which have carbon-carbon closed chain.

Ex : Cyclopropane : C_3H_6 Cyclohexane : C_6H_{12}



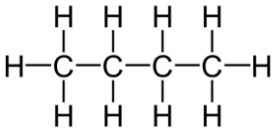


Benzene : It is an aromatic hydrocarbon which has the molecular formula C_6H_6. It has alternating carbon - carbon single and double bonds.

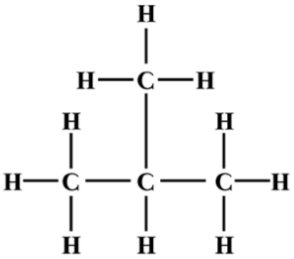


Isomerism : The compounds which possess the same molecular formula but different structural formulae, are called isomers, and the phenomenon is known as **isomerism.**

Isomers of Butane :

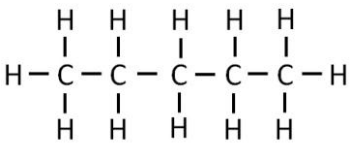


n-butane

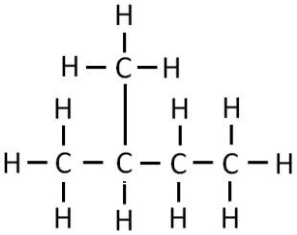


iso-butane

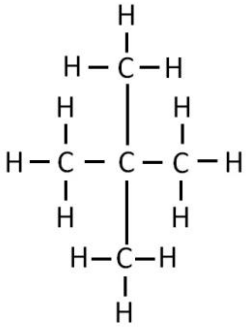
Isomers of Pentane :



n-pentane



iso-pentane



neo-pentane

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Homologous Series :

A series of organic compounds in which every succeeding member differs from the previous one by – CH_2 or 14 a.m.u. is called homologous series.

The molecular formula of all the members of a homologous series can be derived from a general formula.

As the molecular mass increases in a series, physical properties of the compounds show a variation, but chemical properties which are determined by a functional group remain the same within a series.

Functional groups

These heteroatoms and the group containing these confer specific properties to the compound, regardless of the length and nature of the carbon chain and hence are called **functional groups.**

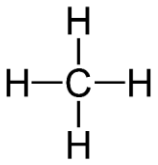
Functional group	Formula of functional group
Halo (Chloro/Bromo)	— Cl, — Br, — I
1. Alcohol	— OH
2. Aldehyde	— CHO
3. Ketone	—C(=O)—
4. Carboxylic acid	—C(=O)—OH

Nomenclature of Carbon Compounds :

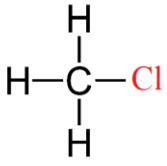
1. Halo alkanes :

If hydrogen atom of a carbon compound is replaced by a chlorine atom, then prefix - chloro has to be used.

Ex :



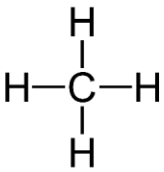
methane



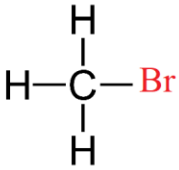
chloromethane

If hydrogen atom of a carbon compound is replaced by a bromine atom, then prefix - bromo has to be used.

Ex :



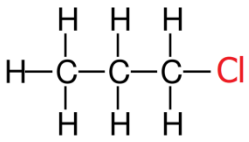
methane



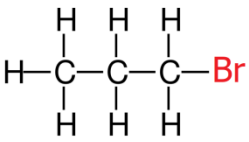
bromomethane

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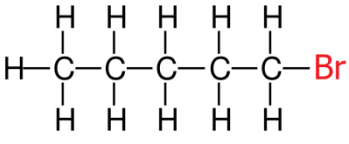
Few more examples :



chloropropane



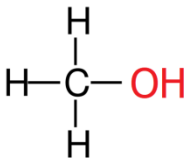
bromopropane



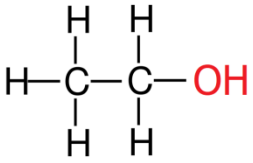
bromopentane

2. Alcohol :

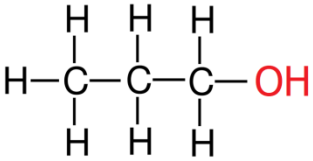
If hydrogen atom of a carbon compound is replaced by **-OH**, then suffix **- al** has to be used.



methanol



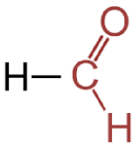
ethanol



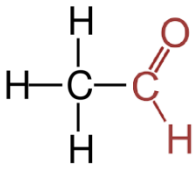
propanol

3. Aldehyde :

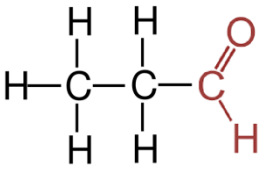
If hydrogen atom of a carbon compound is replaced by **-CHO**, then suffix **- al** has to be used.



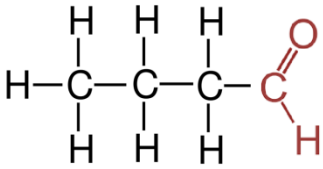
methanal



ethanal



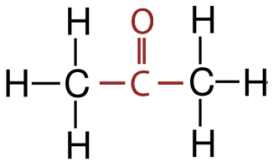
propanal



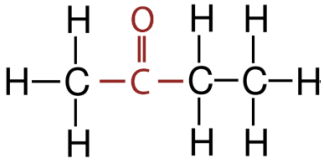
butanal

4. Ketone :

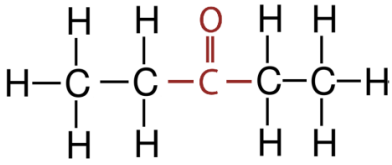
If hydrogen atom of a carbon compound is replaced by  then suffix **- one** has to be used.



propanone

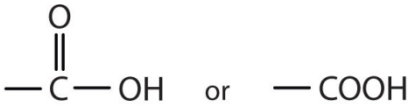


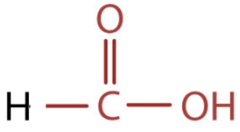
butanone



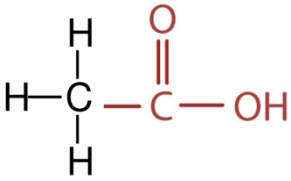
pentanone

5. Carboxylic acids :

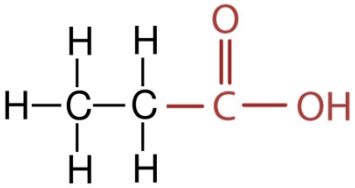
If hydrogen atom of a carbon compound is replaced by  then suffix **- oic acid** has to be used.



methanoic acid



ethanoic acid



propanoic acid

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Chemical properties of Carbon compounds

(a) Combustion:

Carbon compounds burn in air to give carbon dioxide, water, heat and light.

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{Heat} + \text{Light}$$

- Carbon and its compounds are used as fuels because they burn in air releasing lot of heat energy.
- Saturated hydrocarbon generally burn in air with blue and non-sooty flame.
- Unsaturated hydrocarbon burns in air with yellow sooty flame because percentage of carbon is higher than saturated hydrocarbon which does not get completely oxidised in air.

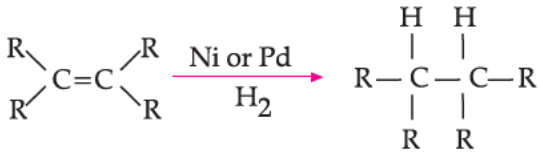
(b) Oxidation:

Alcohols can be converted into carboxylic acid in the presence of oxidising agent like alkaline $KMnO_4$ (potassium permanganate) or acidic potassium dichromate $K_2Cr_2O_7$.

$$\underset{\text{Ethanol}}{CH_3CH_2OH} \xrightarrow[\text{Acidic } K_2Cr_2O_7]{\text{Alkaline } KMnO_4 \text{ Or}} \underset{\text{Ethanoic acid}}{CH_3COOH}$$

(c) Addition Reaction :

Unsaturated hydrocarbons (alkene, alkyne) undergo addition reactions.



In unsaturated hydrocarbon, hydrogen is added in the presence of catalyst palladium or nickel. Vegetable oils are converted into vegetable ghee using this process. It is also called hydrogenation of vegetable oils. Animal fats generally contain saturated fatty acids which are said to be harmful for health. Oils containing unsaturated fatty acids should be chosen for cooking.
(Catalysts are substances that cause a reaction to occur or proceed at a different rate)

(d) Substitution Reaction :

Saturated hydrocarbons undergo substitution reaction in the presence of sunlight.

Ex : Chlorine can replace the hydrogen atoms methane one by one in presence of sunlight.

$$CH_4 + Cl_2 \xrightarrow{Sunlight} \underbrace{CH_3Cl}_{chloromethane} + HCl$$

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Difference between saturated and unsaturated hydrocarbons :

Saturated hydrocarbons	Unsaturated hydrocarbons
Single bond between carbon atoms	Double or triple bond between carbon atoms
Less reactive	More reactive
Generally gives clean blue flame	Burns with yellow sooty flame
Undergoes substitution reaction	Undergoes addition reaction

Ethanol

Ethanol is commonly known as alcohol. It is the second member of the alcohol series. The molecular formula of ethanol is C_2H_5OH. It is a good solvent.

Chemical Properties :

(i) Reaction with sodium : Formation of sodium ethoxide and hydrogen.

$$2CH_3CH_2OH + 2Na \rightarrow 2CH_3CH_2ONa + H_2$$

(ii) Reaction to give unsaturated hydrocarbon : Heating ethanol with excess concentrated sulphuric acid results in the dehydration of ethanol to give ethene –

$$CH_3CH_2OH \xrightarrow{Conc\ H_2SO_4} CH_2 = CH_2 + H_2O$$

In this reaction sulphuric acid acts as dehydrating agent which removes the water from the alcohol

USES :

- It is the active ingredient of all alcoholic drinks.
- it is also used in medicines such as tincture iodine, cough syrups, and many tonics.

Ethanoic acid

The common name of ethanoic acid is acetic acid and it belongs to the group of acids called carboxylic acid. It is the second member of the series.

- The molecular formula of the compound is CH_3COOH.
- 5-8 % solution of acetic acid in water is called Vinegar
- The melting point of pure ethanoic acid is very low hence it often freezes during winter in cold climates. This gave rise to its name glacial acetic acid.

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Chemical Properties of ethanoic acid :

(i) Esterification reaction: Esters are most commonly formed by reaction of an acid and an alcohol. Ethanoic acid reacts with absolute ethanol in the presence of an acid catalyst to give an ester

$$\underset{\text{Ethanoic acid}}{CH_3COOH} + \underset{\text{Ethanol}}{CH_3CH_2OH} \xrightarrow{\text{Acid}} \underset{\text{Ester}}{CH_3 - C(=O) - CH_2 - CH_3} + H_2O$$

Esters are sweet-smelling substances. These are used in making perfumes and as flavouring agents.

Saponification reaction:

Esters on treating with strong alkali like sodium hydroxide they converted back to alcohol and sodium salts of carboxylic acids. This reaction is called as saponification. Because, this reaction is used in making soaps. Soaps are sodium or potassium salts of long chain carboxylic acid.

$$CH_3COOC_2H_5 \xrightarrow{NaOH} C_2H_5OH + CH_3COONa$$

Reaction carbonates and hydrogen carbonates :

Ethanoic acid reacts with carbonates and hydrogencarbonates to give rise to a salt, carbon dioxide and water.

$$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$$

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$$

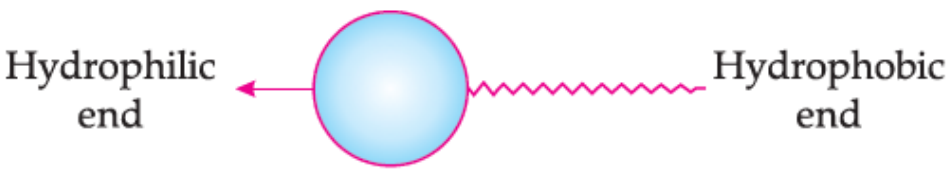
Soaps and detergents

Soaps are sodium or potassium salts of long chain carboxylic acid.

Soap molecule has:

(i) Ionic (hydrophilic) part

(ii) Long hydrocarbon chain (hydrophobic) part



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Cleansing Action of Soap:

Most dirt is oily in nature. The hydrophobic end of soap molecule attaches itself with dirt and the ionic end is surrounded with molecule of water. This result in formation of a radial structure called micelles. Soap micelles helps to dissolve dirt and grease in water and cloth gets cleaned.

- Soaps are effective only in soft water.
- The magnesium and calcium salt present in hard water reacts with soap molecule to form insoluble product called scum. This scum create difficulty in cleansing action.

Detergents are ammonium or sulphonate salt of long chain of carboxylic acid.

- Detergents are effective in both hard and soft water.
- By use of detergent, insoluble scum is not formed with hard water and clothes get cleaned effectively.

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